Digital Twin Architecture

From Theory to Implementation

5 Layers · Modular · Production-Ready

Gian Marco Paldino · 2026 · ULB — Machine Learning Group

About Me

Gian Marco Paldino

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Research Focus

Multivariate Time Series: forecasting, classification, causal inference.
Digital twins for energy, traffic, manufacturing, and agriculture.

Education

Ph.D. — Université Libre de Bruxelles
M.Sc. Summa Cum Laude — Politecnico di Milano
M.Sc. Magna Cum Laude — Université Libre de Bruxelles

Metrics

274 citations • h-index 8 • 14+ publications in Q1/Q2 journals

Technical Expertise

Time Series | Causal Inference | Deep Learning | MLOps | Digital Twins

A Layered Digital Twin Architecture

Five independent, composable layers — each replaceable without touching the others

1	**Generation**	Sensors · Simulators · Synthetic models
2	**Streaming**	Kafka · MQTT · Kinesis
3	**Storage**	TimescaleDB · InfluxDB · PostgreSQL
4	**Processing**	scikit-learn · MLflow · Spark
5	**UX**	React · Leaflet.js · Grafana

Applied in: TrafficTwin • Hand Gesture Recognition • VPP • Plant Health

Now = coherent layering, but different components per project
Objective = **fully uniform digital twin architecture**

Same Layers, Different Components

How each project instantiates the five-layer architecture

	TrafficTwin	Hand Gesture	VPP	Plant Health
1 Generation	SUMO simulation	MindRove EMG armband	pvlib / windpowerlib synthetic models	IoT sensors env + plant health
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2 Streaming	Kafka	BLE stream real-time EMG	Kafka 4 topics	Kafka sensor topics
↓				
3 Storage	TimescaleDB	In-memory sliding window	TimescaleDB hypertables	InfluxDB time-series
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4 Processing	GNN traffic prediction	ML regression joint angles	ML + LP forecast & optimize	ML anomaly detection
↓				
5 UX	React + Leaflet traffic map	Leap Hand robotic control	React + Leaflet VPP dashboard	Dashboard monitoring UI